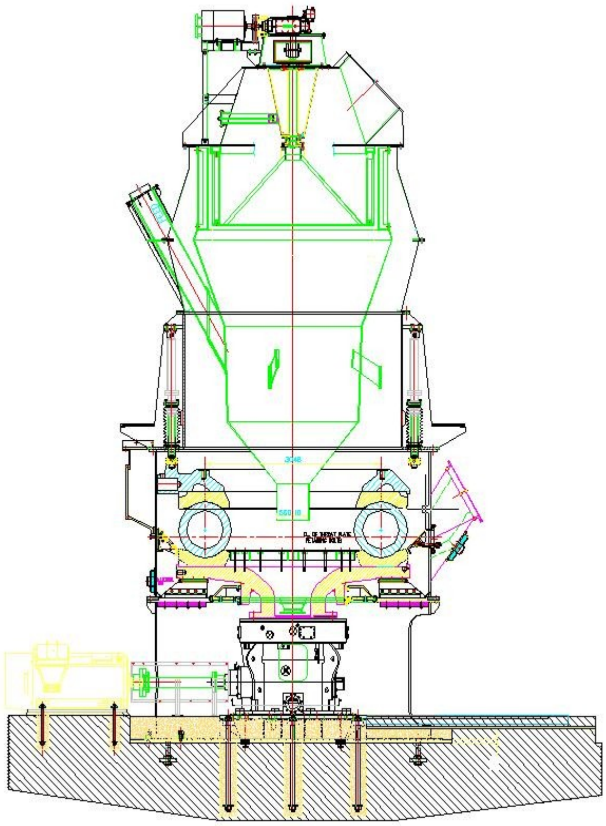
OPERATION & MAINTENANCE MANUAL FOR
BRM 120(10)

Supplied to,
M/s. ACL DARLAGHAT.



Contract No. ECM-000255

India Limited,
SHAHABAD.

INDEX

1.0.0 GENERAL INFORMATION

This instruction manual has been prepared to serve as a guide in operating and maintaining the specific equipment for which it was issued. It is not intended to cover all possible variations in equipment or to provide solutions to specific problems, which may arise. Should additional information be required, ALSTOM India Limited of our field representatives should be contacted.

It must be recognized that no amount of written instructions can replace intelligent thinking and reasoning on the part of the operators and maintenance personnel, especially when coping with unforeseen operating conditions.

It is the responsibility of the operators and maintenance personnel to become thoroughly familiar with the specific equipment covered in this manual along with the pertinent control and auxiliary equipment, to ensure proper functioning, satisfactory performance and personnel safety.

Maintenance, operation, and performance of any equipment not specifically covered in this manual are the sole responsibility of the plant maintenance and operating personnel.

2.0.0 TECHNICAL DATA

2.1.0 MATERIAL CHARACTERISTICS FOR DESIGN

1	Material	:	Pet coke	Coal
2	Feed size	:	25 mm	50 mm
3	Moisture	:	-8% maximum	7-13% maximum4
4	Grindability	:	35 HGI	45 HGI
5	Feed material temperature	:	Ambient	
6	Product fineness	:	1.2 – 2 % R on 90µ	15 – 17 % R on 90µ
7	Feed moisture	:	≤ 1%	≤ 2.5%

2.2.0 MILL – PROCESS PARAMETERS

1	Type of mill	:	Ball race mill	
2	Size and model of mill	:	BRM Vertical 120(10)	
2	Capacity of mill	:	17MTPH–Pet coke	35 MTPH – Coal (Indicative)
3	Product fineness :	:	1.2 – 2 % R on 90µ	15 – 17 % R on 90µ
4	Pressure drop across the mill	:	675 mmwg max.	
5	Table rpm	:	33 - Normal	

2.3.0 EQUIPMENT DATA

1	Grinding elements :		
	Lower ring • Diameter • Height • Material • Surface hardness	: : : : :	 3360 mm 288.5 mm Ni-hard 500-600 BHN
	Upper ring • Diameter • Height • Material • Surface hardness	: : : : :	 3354 mm 288.5 mm Ni- hard 500-600 BHN
	Grinding ball • Type of ball • No. of standard balls • Weight of each ball • No. of fill in balls • Weight of fill in ball • Material of construction	: : : : : :	 Hollow – Spherical 10 nos. 2200 Kg. 1 no. 1452 Kg. Hi- chrome
2	Mill housing support and gear base plate	:	Manufactured from tested quality plates with beams and stress relieved.
3	Mill housing	:	Manufactured from tested quality plates and stress relieved.
4	Throat ring	:	fixed type cast SG Iron castings made in segments
5	Loading unit with Panel	:	M/s TAH MAKE
6	Lower classifier housing	:	Manufactured from tested quality plates.
7	Dynamic classifier with drive without motor, feed chute cone and grit cone.	:	M/s. LNV- Chennai Supply. Motor : Supply by ABB
8	Mill drive detail :		
	Motor (Customer Scope – BHEL make)	:	650 kW , 991 rpm , Sq. cage induction Motor, M/s. BHEL Make

	Coupling	:	GEAR COUPLING : SIZE ET4500 Make : WELMANN Weight: 200 Kg.
	Gearbox	:	Horizontal input & vertical out put - combined bevel helical gear unit with table top coupling half. • Make : M/s ELECON • Reduction ratio: 29.5:1 • Weight: 14,520 Kg. • Oil quantity: 1200 lts (tank), 200 lts (bearing) and 300 lts (pipeline).. • Oil quality: ISO-VG-320 • Lubrication: Forced lubrication system

3.0.0 INTRODUCTION

Vertical ball race coal mill system is an air swept mill system. Ground coal is taken out from mill by suction. These systems can be used for direct firing system or indirect system depending upon the method of application adopted.

The crushed coal is supplied to raw coal hopper by belt conveyor. Then the coal is fed to the ball race mill by means of Rotary Feeder /Drag link feeder of volumetric type. Rod cum cut of gate is provided at the discharge of raw coal hopper to regulate the flow to Feeder.

The raw coal enters the mill and reaches to the center of yoke plate. It spreads and enters between large size ball and rotating table like bottom ring.

The mill is driven by a motor and bevel helical gearbox combination. The motor and gearbox is connected through resilient coupling. The bottom grinding ring and yoke plate get assembled on the gearbox output flange coupling.

The top grinding ring is provided to apply uniform pressure on the spherical balls. The uniform pressure is obtained by hydro-pneumatic system of loading. It consists of 10 nos. hydro-pneumatic cylinders and a panel for pressuring the cylinders. The pressuring medium is Nitrogen gas and lubricating oil is used for sealing the gas.

The hydro pneumatic control panel controls the system provided for uniform loading of the spherical balls. The system automatically compensates for the wear on the balls and grinding rings and provides a constant load at all times. The spider with upper ring moves down slowly as wear of grinding elements takes place.

This ball race mill has built-in dynamic classifier for separation of coarse particles from fines. The dynamic classifier assists to control the fineness of coal.

Hot primary air from is drawn and fed in to the mill housing at two locations and enters tangentially in to the mill, which is used for drying of raw coal, which has higher moisture is the unique feature of the vertical type mill.

Hot primary air enters in the grinding zone of the mill through adjustable throat plate with high velocity. It carries the ground coal of fines and course together to built-in dynamic classifier. In classifier they were separated.

Hard and un-grind able coal or foreign material, which is entering with raw coal feed, will be removed from the mill through the counter weight operated relief gate.

The brush plough bolted to yoke plate. It rotates with yoke and sweeps and drops the hard and un-grind able coal or foreign material in the pyrite box. This reject material will be removed periodically through cut off gate and pyrite box.

4.0.0 DESCRIPTION OF MILL COMPONENTS

Ball race (BRM) 120(10) type coal mill is rugged mill and works without trouble when erected and maintained according to the instructions mentioned in the erection, operation and maintenance manual. The essential features of the BR mill are:

4.1.1 Load bearing packers are provided under the beams, adjacent to foundation bolts, which are embedded in the RCC. The packer centerlines coincide with beam centerline.

Refer drawing no C-BR-1217 for position of load bearing packers for mill housing support and gear baseplate.

4.2.0 Gear baseplate welded to mill housing support dispatched as two different piece. Gearbox housed in the mill housing support and it rests on the gear base plate. The gear baseplate fabricated from tested quality steel plates of thick sections and reinforced with beams. Mill housing support of welded construction rolled from tested quality steel plates, after all welding stress relieved before machining.

Refer drawing no D-BR-1179 for mill housing support and gear base plate drawing no D-BR-1202.

4.3.0 Combined bevel helical gearbox mounted on the machined pads of baseplate with fasteners and located in base plate with provision for movement during alignment. Yoke is assembled with tabletop coupling of gearbox with key and fasteners. External forced lubrication system is provided for gearbox lubrication. Temperature from oil is removed by water circulation.

Refer Elecon drawing no. 6003864-C for gearbox general arrangement.

4.4.0 Mill housing fabricated from tested quality plates and stress relieved after full welding. Bottom flange of housing bolted to mill housing support flange and top flange is bolted with lower classifier housing. Throat ring segments are attached inside of mill housing. Cast spider guides are fixed to mill housing. 3 nos. of inspection doors, 1 no of relief gate, 1 no of ball access door and 4 nos. of spider guides housing are mounted on the outside of mill housing. Two nos. of inlet primary air ducts are connected to it.

4.5.0 Cast steel yoke is keyed and bolted to tabletop coupling of gearbox. Scrapers are attached to yoke to sweep the rejects and drops in to the pyrite box.

4.6.0 Grinding elements consists of top grinding ring, bottom grinding ring and grinding balls. Bottom ring rests on the yoke and which is keyed to it for rotation and top ring is attached to spider and it is keyed to it and stationary one. Grinding balls are inserted between top and bottom tracks of the grinding rings.

4.7.0 Cast spider holds the top grinding ring in position and prevents its rotation. The spider arms are attached to spider and arms are fixed at four locations in the spider guide. In turn the spider guide is bolted to mill housing. The spider will comes down as the grinding elements wear.

4.8.0 Classifier housing mounted on the mill housing. Hydro-pneumatic cylinders 10 nos. are attached at the top flange of lower classifier housing. At the bottom it is bolted to mill housing and spider guides. Coal feed chute is attached to it. Grit cone or reject cone is fixed to stationary part of classifier at the center of mill. All these are fabricated parts.

Refer LNV drawing no. CLF1311638 for classifier general arrangement drawing with feed chute, classifier housings, rotor assembly, classifier roof casing drive arrangement and grit cone etc.

Refer and follow the LNV manual for erection, operation and maintenance of dynamic classifier manual.

5.0.0 PREPARATION FOR ERECTION

The following details shall be studied thoroughly with reference to the individual contract regarding mill type, size, range, application and scope of supply before start of the erection activity.

The position and direction of the mill with reference to datum line to be established and confirmation to be obtained from civil engineering department. Mill and its drive motor centerline position to be checked and confirmed with relative to datum line and it is to be recorded.

1. General arrangement drawing of BR mill.
2. Mill foundation detail drawing with loads.
3. Plant layout drawing and handing
4. Mill shipping assemblies and its weights
5. Scope of supply: Works / Contractor / Customer.
6. Shipping lists with connected Erection drawings.

All the dispatch able units received at site shall be properly identified with respect to document, drawing and mark number. Any short supply or damaged item shall be immediately intimated to the concerned agencies. Critical components such as bearings and motors shall be stored under temporary cover to protect against severe weather conditions.

The exposed machined surfaces and threaded portions leave the factory with protective preservatives. During site storage these preservative coatings shall be reestablished, if necessary.

To start with erection work, proper planning is to be made for tools, tackles, rigging and handling facilities required for the installation of mill. Special attention may be paid to the size, shape and weight of the components as well as the space limitations while proceeding with the installation.

It is to be ensured that the foundation fasteners, sleeves and connected materials are available at site prior/during to casting of RCC pedestal for mill.

After the concrete pedestal is cast and set, visually inspect the foundation for strength and soundness of concrete. Check for cracks, holes, exposed steel reinforcement etc. Especially pay attention to check for any leftover wooden shuttering materials.

RCC foundation checking to be done with respect to the axis of the plant layout (plan) drawings. The position of mill gear base and mill housing support and motor foundation pockets should be checked. Elevation of foundation should be checked with respect to predetermined level from the support columns. Chipping of concrete shall be done, if necessary to correct the elevations.

Permissible tolerances on the foundation

Area	Tolerance
Tolerance in the elevation of concrete foundation with respect to drawing	± 10 mm
Tolerance in center line of foundation pockets with respect to foundation center line	± 2 mm
Tolerance in the diagonals of foundation pocket centers	± 3 mm
Tolerance in the depth of foundation pockets	± 10 mm
Grouting margin to be maintained between mounting plate and concrete foundation.	
Minimum	50 mm
Maximum	150 mm

6.0.0 GROUTING INSTRUCTIONS

This instruction covers materials and methods that shall be used for grouting structural steel and machine base plates.

The space provided for grouting shall not be less than 50 mm and not more than 150 mm.

6.1.0 GROUTING MATERIALS

The preferred grouting material shall be a Standard free flowable, non-shrinkable, high strength cementitious precisian grout like **CONBEXTRA GP2** or equivalent grout mortar. (It is general purpose grout. It is specifically used for cement and resin based products to fill voids beneath load bearing sole plates together with anchoring systems to install permanent or removable anchor and fixing.) This should be a ready-to-use grout and it should be flowable, pumpable or damp pack consistency to produce a hardened grout completely free of both vertical and horizontal shrinkage. Since the grout requires only the addition of water, the required consistency is easily controlled.

CONBEXTRA GP2 is supplied as a ready to use dry powder requiring only the addition of water to produce a free flowing, non-shrink grout which will maintain its original volume during pouring. The material is a mixture of specially processed cement with carefully graded fine aggregate. Additives impart controlled expansion characteristics and reduce the water necessary to produce fluidity. The very low water/cement ratio ensures very high early strength, particularly at low temperatures. The aggregate grading is designed to aid uniform mixing, minimise segregation and bleeding whilst assisting the flow characteristics.

ACC’s anti shrink compound **“SHRINKOMP”** has been found to give satisfactory services for this application. Sand and cement grout may be used as an alternate, but not preferred. Consistency of this material is important for proper placement.

NOTE: FULL CEMENT GROUT IS PROHIBITED.

6.2.0 FOUNDATION PREPARATION

- All loose concrete and defective areas shall be removed from foundation by brush, hammering, chipping or any other means. Surface of foundation must be rough.
- Surface of foundation and base plate must be free of oil, grease and dirt.
- Surface of the foundation shall be well saturated with water for several hours prior to grouting (depending upon the grout mortar).

6.3.0 MIXING AND APPLICATION OF GROUT

- Mixing of grout should take place as near as possible to the equipment to be grouted. All necessary tools and materials should be available to mix and handle grout.
- For producing grout of given consistency, paddle type mortar mixers or revolving drum concrete mixer or any suitable mechanical mixer can be used.

NOTE: DO NOT MIX THE GROUT WITH HAND.

- The dry GROUT MORTAR should be ready to use as it comes from the package and only require mixing with clean water. The quantity of water depends upon the consistency of grout required.

- Grout materials shall be re-tempered with water once mixed.

- Temperature of material is important. A temperature of 30°C material will set quickly: below 30°C material will set slowly. Strength gain will affect by temperature.
- For most base plate applications, place grouts from one side only to assure complete filling of the space to be grouted. Special care should be taken to ensure that there are no air pockets especially under base plate. Use vibrator if necessary.

6.4.0 SHIMS AND WEDGES

i) All wooden shims and wedges shall be removed.

ii) Steel shims and wedges may be left in place, provided they do not protrude through the grout.

6.5.0 IMPORTANT NOTES

- GROUT Manufacturer's published data on mixing grout material shall be followed in order to obtain desired results. Read and understand the material safety data sheets for health and hazards.
- Follow our equipment load data sheets if similar grout material is not available as recommended.

- Grout placement in or around anchor/founation bolts, base plate shall be governed by contract drawings and / or vendors instructions.

7.0.0 ERECTION INSTRUCTIONS

Recommended procedure of erection of mill components is given in this manual. The instructions cover items such as packing steel pads, base frames, bolt tightening procedure, foundation requirements and other information which is pertinent to erection of Mill.

Unless otherwise specified all the components are to be leveled and aligned within 0.2 mm/m.

All parts must be thoroughly cleaned. Dissemble the sub-assemblies wherever necessary; in order to do a thorough job. Machined surfaces have a corrosion protective coating applied at the factory before dispatch. These parts should be thoroughly cleaned off the coating before erection. Special attention should be given to machined surfaces which are likely to get damaged during shipment / transportation or handling.

7.1.0 USE OF PART NUMBERS

Part number in the drawings identifies the equipment parts. They are mentioned in the erection parts list with its description. The sub-assembly drawing pertaining to the part is also mentioned in the erection parts list.

7.2.0 INSPECTION UPON ARRIVAL AT PROJECT SITE

As soon as the equipment arrives at its destination, it should be carefully inspected to determine any shortage or damages. Check each item with the dispatch particulars.

7.3.0 PROTECTION

The ALSTOM India Limited (AIL) machinery is painted and exposed machined surface with covered with suitable protective coating before dispatch.

If after the receipt of equipment at the site, the erection is likely to be delayed, then apply additional protective coating wherever necessary.

Painted surfaces are to be touched up with original paint or similar paint, which will be compatible with the final finish to be used.

7.4.0 STORAGE OF PARTS

All the fabricated parts and steel castings can be stored in an open area, if they have to be erected within short time. Otherwise they should be properly protected from rain so that the mating faces do not get rusted, especially the gearbox mounting pads on the base frame.

Small castings, sealing rings, loading units and other machined components should be stored in covered area.

The storage site selected should be a high level ground with good drainage and free from standing water and mud. Equipment and parts to be stored should be covered properly to avoid damage to the equipment due to corrosion.

The arrangement of the storage should be so arranged, that all parts are easily accessible.
Provide adequate supervision when unloading the parts and also during their reloading for further movement to the place of erection.

Place machinery parts on cradles, wooden sleepers or rails off the ground in a place separated from the active construction area to avoid on site damage. Equipment having machined parts should be covered with the polythene sheeting and /or tarpaulin.

Assembled-machined components, electrical equipment, motors, welding rods, refractories, bolts, nuts and other hardware etc. should be stored in a covered and dry place.

Remove top covers of the bearing block and inspect the bearing and pack with grease is compulsory during erection of the supplied equipment.

Rotors of the electric motors, fans, blowers, gearbox components having assembled anti-friction bearings, should be rotated periodically while in storage and also during the period before plant’s start up.

7.5.0 STORAGE OF ERECTED EQUIPMENT

Drives, motors and bearings are to be lubricated to prevent corrosion. Bearings should be rotated periodically to lubricate all the bearing parts. Apply light coat of multipurpose grease to all threaded portions of bolts to prevent corrosion.

CAUTION: NO MACHINERY SHOULD BE ROTATED DRY.

7.6.0 STORAGE OF RUBBER LINED EQUIPMENT AND RUBBER PRODUCT

Rubber lined equipment and rubber products must be stored away from sunlight, oily place and outdoor seasonal weathering. The storage area should be well ventilated.

Couplings and other seals of gear box

Regardless of storage method, this material must be inspected often to ensure that it is not damaged.

7.7.0 FIELD LIFTED AND HANDLING OF EQUIPMENT

Experienced personnel should supervise the unloading and handling of heavy machinery. Primary considerations are the safety of those working near the machine and the avoidance of the damage to machine itself.

Overhead cranes, crawler cranes, derricks, winches are the best means of lifting heavy machinery. Where they are not available, it is necessary to move the machine by such means as jacks crowbars and rollers. To avoid strains on any part of the machine, the weight must be evenly distributed over as many points as possible.

It is very important that the lifting lugs that are provided should be used properly for which it was meant. Lift only that part, which has the lifting lug, attached to it.

In some instances tapped holes are provided. Their eyebolts are to be used for lifting i.e. top gearbox casing, spider, and yoke.

8.0.0 SEQUENCE AND PROCEDURE OF ERECTION

Before start of the erection check the foundations and obtain the clearance from civil department. First of all check that the place, in and around, where the erection is to be done is cleared of all civil works/debris. The foundation pockets should be checked for its cleanliness.

As per layout drawing and foundation drawing, the heights of foundations, center distance of foundations, depth of foundation pockets and condition of foundation bolts should be checked. The size of foundation pockets should be checked with respect to foundation bolt to give sufficient adjustment during erection.

With respect to centerline of mill and mill drive motor draw the centerlines of base plate bottom beams on the RCC block. Establish the position of load bearing packers on that beam centerline as per drawing. Chip the RCC locally for insertion of load bearing packers and which are to be embedded in the RCC block after leveling. For position of packer plates refer drawing no C-BR-1217.

As per the elevation of gear base plate from the drawing decides the sets of packers. Thickest packer plate to be at the bottom most and thinnest to be at the top most position Thicker packer rests on the embedded load-bearing packer and it is to be blue matched. (Allowed Maximum thickness of packing should be 75 mm).

Two sets, of foundation bolts are supplied, one set of 8 nos. with shorter length for gear base plate and another one set of 8 nos. with longer length for mill housing support. Ensure the correct length of foundation bolts at correct foundation pockets.

8.1.0 GEAR BASEPLATE AND MILL HOUSING SUPPORT:

Lift the gear base plate to suitable height. Hang the eight nos. foundation bolts for base plate and similarly lift the mill housing support and hand the other 8 nos. of foundation bolts to the stool of the mill housing support with their nuts without spring washers at this stage. This will ensure that the bolts hang vertically down. Allow sufficient length of the bolts to project above the nuts to accommodate the spring washers.

Set the mill housing support down on the three jacking screws, using a steel thrust plate under each jacking screw. Guide the foundation bolts into the pockets in the concrete foundation block. Check the correctness of base plate position with respect to centerlines mill and drive motor.

Wedge the mill housing support and gear base plate against the walls of the recess in the concrete foundation block so that it cannot disturbed accidentally.

By means of the three jacking screws and a Dumpy level, set the base plate to the correct level, i.e. top of withdrawal trolley rails relative to floor datum level.

The grouting is to be made in two stages, one half grout of foundation bolts and another full grout with mill housing support with gear base plate, set of packers and foundation bolts.

Grout all foundation bolts with non-shrink cement up to 40 mm from top of the foundation pockets. Ensure that the bolts will remain vertical. Use vibrator or any other systems to remove are air pockets in the grout. Allow minimum 7 days for grout to set and achieve sufficient strength to hold the foundation bolts when they are tightened.

Remove nuts from the foundation bolts after setting of grout. First check the level of the base plate and adjust where necessary if required by means of the jacking screws and using the Dumpy level.

Make sets of steel packers comprising of various thickness plates, including the thinnest available and place them under the base plate and mill housing support bottom flange. The packers are to be provided both sides of each foundation bolt and they are to be placed as close to the bolts as practicable. For position of packers refer drawing. Fit spring washers and nuts to the foundation bolts. Turn back the jacking screws from their thrust plates to ensure that they do not affect the final adjustment of level. Tighten the nuts of the foundation bolts to pull the base plate and mill housing support hard down against on to the packers.

Check the differences in levels between the faces of the gearbox resting pads and support stools. (By using a micro-level suitable length, straight edge in conjunction with an engineer's precision level and feeler gauges). Record the level differences of pads and stools. Determine the sequence and degree of adjustment to the thickness of packers required to bring the base plate and support within the permissible limits of 0.15 mm total between the levels of all pads.
The gearbox is supported on 12 nos. of holding down bolt pads.
To achieve the required levels of pads add or remove the packers as required. The packer may be either single or a set depending upon the difference in the level. Again tighten the base plate and support hard down on the packers. This exercise may be required several times

before the required result is achieved. During this process, the jacking screws can be used for lifting and lowering of the base plate and support. The nut of the foundation bolts to be loosened for adding or removing packers. The jacking screws must be turned back from their thrust plates every time before the nuts on the foundation bolts are to be tightened down.

By the above method if it is not possible and practicable to achieve the required pad levels differences within acceptable limits then dress the pads by grinding the high spots.

When the level of all pads and stools are as close to the permissible tolerance as can be practically achieved by the method of adjusting the packers, check that al the packers set is in solid in response to hammer blow. This indicates that they are tight.

Remove and discard the three jacking screws and their thrust plates.

Re-check the position of the base plate and housing support, with respect to the datum line, mill centerline and motor centerlines, to ensure that it has not been accidentally moved, and check the height of the baseplate relative to the floor datum level.

Check and clean the center portion and top plate. Remove any damage if occurred to center portion and inner face of the top plate.

Check the concentricity of the top seal air ring and the vertical gap between a straight edge and the shoulder of the sleeve of the aligning post/PLUMB.

By means of the jacking screws, adjust the height and level of the machined top surface of the seal air casting as necessary.

With the aid of jacks placed at strategic points, adjust the position of the support, as necessary, so that the air seal housing ring is concentric within the permissible tolerance of 0.25mm.

Re-check the height and level of the machined top surface of the seal air casting and adjust, as necessary, by varying the thickness of the sets of packers under the bottom flange of the support.

Check the concentricity of the air seal housing and adjust, as necessary, by loosening the bolts which attach the seal air casting to the mill housing support top plate and centralising the seal air casting by means of the four adjusting screws. Re-tighten the attachment bolts.

*Holes check up, gear box rotation by hand

Check that the openings in the support are in their correct positions, circumferentially, relative to the baseplate.

Grout the base plate with housing support with non-shrink cement. Grout under the baseplate including transverse beams up to 5 mm below the grouting pockets. Ensure that there are no air pockets in the grout and grout fully filled at all area. Air releasing holes also provided at different locations. Allow the Cement to harden and achieve sufficient strength for the subsequent erecting operations.

Grout up to just above the bottom level of the flange, filling the remainder of the foundation bolt holes in the process. Ensure that there are no air pockets in the cement that the spaces are completely filled and the cement extends to some extent beyond the diameter of the flange, both inside and outside. Allow the grout cement to set and harden.

Position the four securing brackets on the top face of the baseplate, adjacent to the jacking screws and hard up against the inside of the mill housing support. Weld the brackets to the baseplate and support, filling the slots with weld and welding all-round the sides of the brackets.

Weld the two gearbox trolley rail support into the bottom flange of the mill housing support. Bolt the gearbox trolley rails loosely to the baseplate and set up the rails to the correct height and level relative to the rails on the baseplate and the floor datum level, by means of adjusting the height of the rail support bars and with packets under the rails. Tighten the bolts attaching the rails to the baseplate. Weld the rail support bars to the flange of the mill housing support.

The space inside the mill housing support, under the baseplate, around the mill housing support and in the floor recess for the gearbox trolley rails can now be filled with concrete, ensuring that all cavities are filled and there are no air pockets in the concrete. The concrete is to be flush with or within 6mm of the top face of the gearbox baseplate.

A square recess of about 22mm width and depth must be provided in the concrete along the top inside edge of each trolley rail, for the full length of the rail, to accommodate the flanges of the gearbox trolley wheels.

When the concrete is set, harden down all the foundation bolts in the gearbox baseplate and the mill housing support.

8.2.0 GEAR BOX.

The gearbox compete must **not be** lifted by means of the eyebolts. These eyebolts are to be used for lifting the top half of the gearbox case only.

a) Before installing the gearbox in the mill housing support, it is necessary to remove the yoke coupling guard if provided.

b) Lift the gearbox, support it on timber blocks, inspect and clean the holding down bolt pads and the centre locating spigot. Ensure correct seating on the gearbox baseplate.

Place the gearbox between the trolley rails close to the mill housing support, keeping the underside clean. Place the gearbox withdrawal trolley on the trolley rails.

c) Clean the faces of the gearbox holding down bolt pads, center pad, and the horseshoe stop plate of baseplate. Ensure that gearbox pads will fit flat on the baseplate. The gearbox will be located correctly in the horseshoe.

d) Raise the gearbox by means of hydraulic jacks placed under the three jacking lugs, two at the input end and one at the other end. Clean the underside of the holding down bolt pads and the centre-locating spigot.

e) Push the trolley under the four support lugs of the gearbox and fit the four locating pins in the holes of the gearbox lugs and the trolley lugs, to secure the gearbox whilst it is being moved. Lower the gearbox on to the trolley and remove the hydraulic jacks.

f) Push the trolley and gearbox into the mill housing support until the centre spigot under the gearbox locates in the baseplate.

g) Raise the gearbox by means of the hydraulic jacks placed under the three lugs, remove the four locating pins, and withdraw the trolley from the mill housing support. Lower the gearbox on to the baseplate and remove the hydraulic jacks.

h) Fit four nos. of gearbox adjusting brackets on the base plate. By means of the two adjusting screws located at the non-drive end of the gearbox, jack the gearbox firmly into the horseshoe of the baseplate. Do not use excessive force on the adjusting screws.

The alignment and adjustment of the gearbox is required when the mill driving motor is installed. The mill gearbox and motor aligned, and the coupling gap is set as required. This alignment of the gearbox will be achieved by means of the two adjusting screws located at the sides of the gearbox, at the input end.

In the meantime, the gearbox alignment should be set approximately to the datum line, by means of these adjusting screws. If it is necessary to make some more adjustment for alignment, turn back the adjusting screws at the non-drive end, to permit the gearbox to be turned around the horseshoe location, as required. When the gearbox is aligned, the adjusting screws at the non-drive end should be tightened to again locate the gearbox firmly in the horseshoe of the baseplate.

Check that the holding down bolt pads of the gearbox and baseplate are in contact.
When the gearbox located and aligned by means of the four adjusting screws in the adjusting brackets, lock these screws in position by means of their locking buts.

The gearbox holding down bolts are now to be fitted and tightened down in accordance with the instruction on the gearbox drawing.

8.3.0 MILL MOTOR.

Refer to Foundation drawing, Mill Motor sole plate drawing, Mill Motor drawing and drawing for coupling between Motor and Gearbox, and operating Instructions for coupling. Check the dimensions of the motor foundation block in relation to the mill foundation, and mark all centerlines in X and Y direction of the motor location on the plinth.

Erect the motor on to its soleplate, using the foundation bolts and spring washers provided.

Lift the motor and soleplate assembly, and hung the foundation bolts to the soleplate. Allow sufficient length of bolt to protrude above the nuts to accommodate the spring washers, which will have to be fitted later.

Lower the motor and soleplate assembly on to the motor foundation block, guiding the foundation bolts into the pockets in the concrete vertically.

Make up 8 sets of steel plate packers, and place them under the soleplate, such that foundation bolt will be between the two sets of packers. Ensure that the bolt pockets are not obscured.

Adjust the thickness of the packers, and place them under the soleplate, so that the gearbox and motor half couplings are in alignment to the requirements of the coupling manufacturer standard. The gap between the coupling half must also be set with the motor shaft positioned at the magnetic center.

It is important at this stage to ensure that the gap and the lateral alignment between the coupling halves are correct. The horizontal alignment can be adjusted by altering the thickness of the soleplate packers after the foundation bolts have been grouted. Grout foundation bolts up to 40 mm below the pocket level with non-shrink cement. Ensure that the bolts remain vertical and that no air pockets exist in the grout cement. Allow the grout to set and gain sufficient strength to hold the foundation bolts when they are tightened.

After the grout has set, remove the nuts from the foundation bolts, and readjust the thickness of the packers under the soleplate so that the motor half-coupling is 7 to 8 mm below the gearbox half-coupling. This is to allow for exchange of motors during maintenance. Packers and shims under the motor feet will make final adjustment.

Replace the spring washers and nuts on the foundation bolts, and recheck the coupling gap and lateral alignment after tightening the nuts. Check and confirm the gap and alignment are correct when foundation bolts are in fully tightened condition, adjust if required. The grouting under the soleplate can be carried out. Grout sole plate and balance portion of foundation pockets with non-shrink cement. Ensuring that there are no air pockets or unfilled spaces under the soleplate.

Allow the grout to set. Then carry out the final alignment of the coupling halves as per manufacturer's standard. All round the circumference of the half-coupling check lateral alignment with slip gauge and filler gauge. Adjust the motor with coupling by using alignment block with adjusting screw to required direction.

Check the horizontal alignment of motor half coupling with dial gauge to the motor half-coupling by rotating the motor shaft, and taking readings at 90°, 180°, 270° and 360°. Adjustments are to be made to the motor and gearbox till the readings all round the circumference of the coupling are in accordance with the manufacturer's tolerances.

The gaps between the motor feet and soleplate should be accurately measured and machined plate packers to be fit under the feet. Each packer should be equal in area to a motor foot, and drilled to suit the motor holding down bolt. The packers should be machined to ensure that motor is 1- 2 mm lower than gearbox.

Fit packers under the motor feet and tighten the holding down bolts. Check lateral alignment and the gap between faces of the half coupling and adjust if necessary. When final, tighten the adjusting screws at the gearbox and secure locking nuts.

Finally, adjust horizontal alignment by inserting shims under the motor feet as required. Tighten the motor holding bolts. Complete the final assembly of the coupling in accordance with the manufacturer's instructions.

After trial run of the motor, fit the coupling guard and drill holes for dowel pins on the soleplate. Use pre-drilled 12mm holes in the motor feet as guides and fit dowel pins.

8.4.0 YOKE ASSEMBLY.

The Yoke plate provided with four nos. of tapped holes for lifting eyebolts. These holes are under cover plate. Remove the cover plate for access to the tapped holes.

Lift the yoke assembly and lower it, with the stem through the seal air rings, on to the gearbox output coupling, with the keyway in line with the key in the coupling. Fit the key to it, as necessary blue match grind or file the key to suit.

Clean the underside of the yoke stem and support it on the on the chocks supplied, placed under the yoke plate.

Before finally lowering the yoke assembly on to the gearbox output coupling, ensure that all surfaces are clean and free from damage. Carefully fit the bore in the underside of the yoke stem on to the spigot of the gearbox output coupling, and settle the yoke assembly to the coupling. Check that the faces are in contact all round. Fit the set pins and locking plates to the yoke stem and the gearbox coupling. (Blue match)

Check the clearance between the yoke stem and the lower seal air ring, for concentricity and adjust, as necessary, by means of the four centralising set pins in the lugs which are incorporated in the bottom flange of the seal air casting.

It will be necessary to loosen the flange bolts of the seal air casting to adjust the concentricity of the seal air rings. Re-tighten the flange bolts when the seal air ring is correctly positioned.

8.5.0 BOTTOM GRINDING RING.

Lift the bottom grinding ring, by means of two special lifting hooks supplied. Support it on timber blocks. Clean the underside faces and the locating bore of the ring.

Clean the face and locating spigot of the yoke plate, and remove any damage, to ensure the ring can locate and seat correctly on the yoke plate.

Lift the bottom ring and lower it on to the yoke. Ensure that the match marks on the ring and yoke are correctly located. The Yoke can be turned by turning the input coupling of the gearbox and match the match marks of the keyways in the bottom ring with the keys in the yoke.

When the ring is firmly seated on the yoke, check that the faces of the ring and yoke plate are in contact. Small gaps, for short circumferencial lengths, are acceptable, but not more than 0.2 mm gap and for lengths not greater than 300 mm. These are to be checked by feeler gauges around the outside circumference of the ring. If any gaps greater than these dimensions occur, the high spots on the face of the ring can be dressed off with a disc grinder. Under no circumstances should the face of the yoke be dressed except to remove any damage high spots.

When checking the seating between the ring and yoke faces, check also the keyways in the ring those are not riding on the top surfaces of the keys in the yoke. This can be checked by inserting long feeler gauge between the top surface of the key and the keyway in the ring. The contact between the keys and keyways

Should be at the sides. If the keyways were riding on the keys, this would hold the ring off the yoke face. If any dressing is required, it must be in the keyway of the ring and not the key.

It is important to clean the faces of the ring and yoke, after any dressing has been carried out, before the faces are brought together for blue matching. The contact should be about 80% at outer landing. The inner landing may be slightly lower or it levels that of outer landing. Level need to be checked(Make a note in Drawing also)

Before working on or cleaning the underside of the ring, support it on timber blocks. As the rings and yokes have been not made trial at the works, before dispatch, dressing should be necessary during at erection.

When the bottom-grinding ring is satisfactorily seated on the yoke, replace the yoke cover plate, which had been removed when installing the yoke plate assembly.

8.6.0 SCRAPER ASSEMBLY.

Fit the scrapers to the yoke plate assembly as indicated. Ensure and maintain 3-mm clearance between the bottom of the plough blade and the mill housing support top plate.

8.7.0 MILL HOUSING.

Before the mill housing is positioned, check that the throat area, which is set at the works, is in accordance with the requirements of the contract.(drawing number to be mentioned/Rough value)

Lift the mill housing assembly and support it on timber blocks. Clean the bottom of the bottom flange and the locating recess diameter and remove any damage, and to ensure correct seating on the mill housing support.

Clean the top face and the locating diameter of the top flange of the mill housing support and remove any damage. Refer throat plate arrangement drawing and check that the packing rope is in position between the throat plate and the bore of the mill housing.

Lift the mill housing assembly; turn it as necessary, to line up the match mark on the bottom flange with the match mark on the top flange of the mill housing support. Check that the primary air inlets are in the correct circumferential positions to suit the plant and mill layout drawings.

Lower the mill housing assembly over the bottom-grinding ring. Keep the housing level, so that the inside diameter of the throat plate does not jam on the bottom-grinding ring.

Carefully locate the recess in the housing bottom flange on to the locating diameter of the top flange of the housing support and settle the mill housing down on to the support.

Check that the faces of the flanges are in contact. If they are not in contact, lift the housing, check and verify the cause and remove any obstructions of fouling areas, then clean the faces of the flanges and lower and housing again.
When the faces are in contact, re-check that the primary air inlets are in the correct circumferential positions to suit the layout drawings.

Lift the housing assembly, apply Plastic Hermatite or similar jointing compound to the top flange of the housing support, sufficient for an air tight joint, lower the housing assembly and bolt the housing and support flanges together.

8.8.0 GRINDING BALLS.

One set 10 nos. of grinding ball is supplied for the mill. These should be checked with caliper or shop markings. These balls are graded in three sizes as large, medium and small and supplied in matched set. There will be variation of diameter. These are to be arranged in necklace pattern from largest one at the center and smallest at 180º opposite. Each ball in a set is marked with a letter L, M or S, indicating the diameter of the ball in relation to the tolerance range on diameter. L indicates the larger, M for middle and S the smaller diameter.

Arrange balls as mentioned and place them in the ball track of the bottom grinding ring, using the special ball lifting tackle.

924 mm dia. To 926 mm dia. – **L**arge

922 mm dia. To 923 mm dia. – **M**iddle

921 mm dia. To 920 mm dia. – **S**mall

8.9.0 SPIDER GUIDE HOUSINGS.

All four spider guide housings are fitted to the mill housing, and the spider is set up inside the mill housing. The appropriate gaps between the leading and trailing wear plates on the spider and spider guide housings are set and all necessary parts are match marked.

The Spider is dispatched independently. Two of the spider guide housings are removed from the mill housing and shipped as loose items, because the complete housing assembly exceeds the permissible maximum width for transport. Some cases all the four nos. of spider guide housings are fit with housing before dispatch from works.

If two nos. were dispatched clean the faces for the loose spider guide housings and of the mill housing and remove any damage, to ensure correct contact of the faces when assembled.

Apply Plastic Hermatite or similar jointing compound to the faces of the mill housing, sufficient for airtight joints.

Lift each spider guide housing, in turn, and fit to the mill housing, checking that the match and identification marks agree.

Ensure that each spider guide housing is fixed firmly against the locating lugs (end stops) at the left hand side of the opening in the mill housing. The top face of the spider guide housing flange is truly level with the top face of the mill housing flange at both sides.

8.10.0 TOP GRINDING RING AND SPIDER ASSEMBLY.

Top grinding ring, Spider, Push rod Cup loading Cylinder and Spider Guide are the connected parts.

- Push rod assembly need to bec hcekced before installing the spider.

CAUTION: The top grinding ring and spider assembly should be fitted together before installing in the mill housing.

Using the four special clamps supplied, set the top grinding ring, flat face uppermost, on timber blocks, which are adequate to support the ring and spider assembly.

Note that each clamp is fitted with a screwed retaining pin which must be used to ensure that the ring does not slip in the clamp whilst being moved.

The spider has three tapped holes in the bosses around the bore, for the fitting of eyebolts for lifting purposes. These eyebolts are the same one, which are used for lifting the yoke
The tapped holes in the spider are fitted with screws, as plugs, to keep the holes and threads clean. Remove these screws and fit the eyebolts.

Lift the spider assembly, support it on timber blocks, clean the underside faces and the locating diameter and remove any damage.

Clean the top faces and the locating bore of the top grinding ring and removes any damage, to ensure the spider can locate and seat correctly on the top grinding ring.

Turn the spider assembly, as necessary, to line up the match marks with those on the top-grinding ring and to line up the keys with the keyways in the ring.

Lower the spider assembly on to the ring and, when it is firmly seated, check that the faces of the spider and ring are in contact and that the keys are not riding on the keyways in the ring.

The instructions under item no.8.5.0–Bottom grinding Ring, relating to the fitting of that ring and the yoke, will also apply to the fitting of the spider and top grinding ring.

If any dressing is necessary, it must be to the ring faces and keyways. Under no circumstances should the faces of the spider or keys to be dressed, except to remove damage high spots.

When the spider assembly is satisfactorily seated on the top grinding ring, remove the eyebolts and replace the screws.

Do not attempt to lift the spider and ring together by means of the eyebolts. Adequate slings are to be used for this operation and protective material placed between the slings and the points of contact with the spider and the ring.

Turn the spider and ring assembly, as necessary, to line up the spider arms with the appropriate spider guide housings, to ensure that the match and identification marks.

Lower the spider and ring assembly into the mill housing, keeping it level to prevent the spider arms jamming in the spider guide housings, and settle the top grinding ring on to the grinding balls.

8.11.0 CLASSIFIER SHELLS.

Lift the lower classifier housing: refer drawing D-BR-1226 and classifier housing with raw coal chute. Refer LNV classifier GA drawing no. CLF13-1.1638 and support it on timber blocks. Clean the underside of the bottom flange and remove any damage. Weld the loose lifting lugs to suit requirements.

Clean the top faces of the top flanges of the mill housing and the spider guide housings and remove any damage to ensure a satisfactory joint with the flange of the classifier housing.

Turn the classifier shell, as necessary, to line up the match mark on the bottom flange with the match mark on the top flange or position the raw coal feed chute to the mill layout drawing.

Apply plastic Hermatite or similar jointing compound to the top faces of flanges of the mill housing and spider guide housings.

Set the classifier shell on to the mill housing, keeping their flanges concentric and ensure that the match marks are aligned and check the position of raw coal chute and position of loading cylinder are aligned and are in correct position to suit the Plant and Mill Layout Drawings. Bolt the flanges together.

Lower housing LNV drawing no. CLF13-1.1657 to be mounted on the classifier housing similar way that of classifier shell and aligning coal chute. Apply plastic Hermatite or similar jointing compound to the top faces of flanges of the classifier shell and bottom cone.

Pack any gaps between the top and bottom cones with soft rope to ensure an effective seal.

8.12.0 PRESSURISED GAS BALL LOADING CYLINDERS & PUSH RODS

The push rods for the ball loading system are handed. This handing refers to the radial position of the swivel pin, which is fitted to the spherical end, relative to the positions of the plain holes in the top flange.

The ten pressurized gas cylinders are all to the same hand but have to be assembled with the push rods to make assemblies.

The bellows seals and retaining clips for the cylinders are shipped separate from the cylinders, to prevent damage during transit. Clean the seals; the flanges of the cylinders and the projecting part of the cylinder rams.

Fit the bellows seals between the cylinder body and the flange of the ram, ensuring that the drain connection in the body is not obstructed.

It is very important that there is no dirt, fluff from cleaning materials or other foreign matter inside the bellows seals or on the ram.

Clean the faces of the flanges and locating diameters of the ten cylinders and ten push rods and remove any damage.

Fit the cylinders and push rods together to form assemblies, as shown on the Arrangement of Cylinder Mounting drawing no. 6309-1-0702-424 and D-BR-1239.

Insert stub pipe, pipe clips and bellows on the push rod assembly. Then fit the distance piece between push rod assembly and cylinder flange, to retain the cylinder ram in the retracted position.

Attach cylinder support brackets to classifier shell as shown in loading unit assembly drawing. Remove the keep plate from the outer end of the anchor pin, in each bracket, and withdraw the pin, taking care to retain the two loose bevel washers for future use.

Clean the support bearing of each cylinder, including the bore, the bores and faces of the support brackets, the anchor pins and the bevel washers.

Fit a cylinder and push rod assembly to each support bracket, as shown on the Arrangement of Cylinder Mounting drawing, with the oil and gas ports facing the angular end of each bracket. Apply a coating of suitable grease to the anchor pin and the bevel washers, before assembly, and ensure that the bevel face of each washer is towards the cylinder bearing.

Replace the keep plate over the end of each anchor pin and screw in the stop pin to restrain the cylinder and push rod assembly.

Fit the clips to the bellows seals to the push rods and stub pipe as shown on the Arrangement of Cylinder Mounting Drawing.

Lift each cylinder and push rod assembly, in turn, lower the push rod through the appropriate cylinder pocket in the classifier housing and re-fit the support bracket in the bridge piece. Ensure that the all the assemblies are fitted properly at correct positions. The gas and oil ports in the cylinders must face outwards in each of cylinders.

Remove the retaining bolts from the flanges of the cylinders and push rods. Lower the push rods and locate the spherical ends in the cups in the spider. Ensure that the swivel pins are securely located in the recesses in the cups. It may facilitate lowering the push rods if the plugs in the gas ports are withdrawn from the cylinders, to break any air lock. If this is done, each cylinder and push rod must be dealt with in turn and the plug replaced as soon as the push rod is located in the cup. It is essential to ensure that no dirt or other foreign matter enters the gas port in the cylinder.

Note: It is recommended that the cylinders be exercised at 6 monthly intervals and therefore they should not be fitted to the mills more than six months before the mills are put into service.

8.13.0 PRESSURISED GAS BALL LOADING SYSTEM

The ball mill loading system cabinet is supplied with all the necessary valves switches etc fully installed. This cabinet is to be mounted on the angle iron plinth and located beyond the mill foundation block as shown on layout drawing.

Refer M/s. TA Hydraulic drawing no U2908 30 00 – 3 for Mill loading cabinet assembly and detail and refer drawing U2908 34 00 – 3 for gas and oil connection for loading unit panel. Also refer the P and ID for details refer drawing number U2908 36 00-3.

The gas bottle guard is to be attached to the cabinet, which has to be drilled at site for the attachment of the guard and the chain for securing the gas bottle.

The gas connection from the bottle to the bulkhead fitting in the cabinet and the oil drain pipe from the drain valve below the sight gauge in the cabinet, are supplied as loose items, for fitting at site.

The piping between the loading cylinders and the bulkhead fittings in the cabinet has to be run at site. The piping is supplied in straight lengths and the pipefitting as loose items. Refer to the Piping Diagram and the Drawings for the cylinders, Cabinet Assembly and Installation. On completion, check for leakages in the system.

8.14.0 GEARBOX LUBRICATING OIL SYSTEM

M/s. Elecon supplied gearbox and its lubrication system. This forced lubrication unit is to be located adjacent to the mill housing support and the driving motor, refer layout.

The pipe lengths are supplied in running meters to be cut at site and bend at site as per layout drawing. The cooling water flow and return pipe work to the oil cooler are to be provided by the client.

Each length of piping is to be connect the oil pump and cooler unit to the gearbox, which is located inside the mill housing support. An elongated slot is provided in the mill housing support, at the gearbox-input end, for the passage of the oil pipes.

It is essential that the bores of the oil pipes are completely clean and free from foreign material and scaling when installed. Therefore, the pipes should be cleaned by flushing before being installed, to remove any scale formed during heating for bending or during the welding of the flanges.

DESCRIPTION AND WORKING OF LUBRICATION SYSTEM

The system consists of a gear pump driven by an electric motor, the oil filter, oil cooler and pressure gauges to show pressure at outlet of pump, outlet of filter and inlet to the gearbox. The system also has dial type thermometer to show the temperature of the oil being supplied to the gearbox. Apart from this, the system has a relief valve, flow switch, oil pressure switch and a set of stop valves and thermo-switches as a safety device. All these are inter-connected and mounted on a base plate.

The above components are connected as shown in the schematic diagram of the lubrication arrangement showing the direction of flow of oil and water in the oil cooler. Refer drawing
6003864 – C general arrangement of the gearbox installation.
There are three flanged terminal connections with the lubricating equipment. These terminals have to be connected to the corresponding terminals on the mill gearbox. The three connections are suction, delivery and relief. When the pressure in the system increases, due to some blockage, the relief valve opens and allows the oil to bypass in to the gearbox sump. When the delivery pipe is starved, the flow switch gives alarm or trips the main mill motor. Thermo-switches provided at the inlet to the pump and gearbox is set to maintain the lube oil temp within limits. The working of the system is simple. As soon as the pump is started, the oil from the gearbox sump sucked and pumped through the oil filter and cooler to the

delivery line. This line has a pressure gauge, pressure switch, a dial thermometer and thermo-switch. In case the oil pressure is not enough, the flow through the circuit falls and the pressure switch opens disconnecting the power to the main motor.

Erection of the lubricating system consists of positioning the lubrication equipment, according to the layout drawing and connecting the three terminals on the lubrication equipment to the three terminals at the gearbox end. As explained earlier, the three connections are delivery line, which supplies the oil to different points inside the gearbox. Suction line, which draws the oil from the gearbox and the relief line, which comes into play, only when there is pressure rise in the system. Three terminals on the gearbox should be identified from the gearbox drawing and connections made accordingly.

The flow switch, temperature switch and the pressure switch should be connected as shown in the sketch. If the pressure falls below 1.0 Kg/cm^2 the pump should change over to stand by pump. The pressure switch should be adjusted such that the mill main motor trips when the lubrication oil of gearbox falls the pressure below 0.7 Kg/cm^2. Normally, the oil pressure should be maintained between 2.5-3.0 Kg/cm^2 on the pump discharge side. The relief valve should be set to open at 5 Kg/cm^2. The temperature of oil will be set between 54°and 70°C. If the temperature of the oil exceeds 70°C the mill main motor will trip. The oil flow shall be maintained of 90 LPM for the smooth operation of Gear Box.

8.15.0 HYDRO-PNEUMATIC PANEL

The hydro-pneumatic control panel controls the system provided for loading the spherical balls. The system automatically compensates for the wear on the balls and the grinding rings and provides a constant load at all times.

DATA

Gas	:	Nitrogen
Oil	:	Shell gear oil SAE 80 or equivalent
Gas working pressure	:	55 - 58 Kg/cm^2
Minimum gas pre-charge pressure	:	70 Kg/cm^2
Full gas pre-charge pressure	:	240 Kg/cm^2
Gas relief valve setting	:	70 Kg/cm^2
Oil pressure	:	67 - 70 Kg/cm^2
Oil relief valve setting	:	80 Kg/cm^2
Pressure switch '1PSLL31204A'	:	22 Kg/cm^2 ("makes " on falling pressure to shut down or alarm the mill motor)
Pressure switch '1PSLL31203A'	:	35 Kg/cm^{2-}("makes " on falling pressure to give alarm)
Bottle capacity	:	7 m^{3-} Approx.

8.15.1 DESCRIPTION

The system may be divided into distinct parts; viz. the **gas side**, which is the operating medium, and the **oil side**, which acts mainly as the lubricant for the cylinder and sealing agent.

Gas is supplied to the system from a Nitrogen gas bottle; the pre-charge gauge indicates the pressure in the bottle. A regulating valve meter the supply, so that the gas always enters the system at the same pressure, provided the bottle contains the gas at a pressure appreciably above the working pressure. The gas relief valve protects the system from excessive pressure.

The panel has 3 phase gauges, which are visible from the outside, the balance of the hydraulic and pneumatic circuit is enclosed in the cabinet. The cabinet has three terminal connections (T). One for connecting the Nitrogen bottle, second is outlet for N_2 mains going to 10 cylinders and third for the oil outlet line, which supplies the oil to the 10 loading cylinders for lubrication of the cylinder and sealing the gas.

In the gas delivery line there are pressure switches are provided. Namely for normal operation, for low pressure, very low pressure and alarm. Instrument set points and operating range are shown in the drawing no. U2908 36 00 – 3 and instrument schedule no. . If the gas pressure falls less than 35 Kg/cm^2 the mill motor stops.

Gas is also supplied to the large side of the accumulator in which there is oil on the annular side. Since the oil pressure equilibrium is maintained by virtue of the position of the piston. The oil is supplied to the accumulator by means of a hand pump that draws oil from the supply tank through a filter and delivers it through a non-return valve. The relief valve ensure that the oil pressure does not exceed 80 Kg/cm^2 and the needle valve in parallel with it allows the pressure to be relieved for maintenance. The drawing shows the valves as they should be in the operating condition.

The oil pressure in the circuit is indicated on the oil pressure gauge on the panel. The pressurised oil acts between two sets of seals pressurising them to make an effective seal to prevent leakage of gas. The oil pressure is always 10 Kg/cm^2 approx. more than the gas pressure. With this condition, the may leak past and enter the gas space and ultimately find its way to the lowest point in the system and accumulate there till the oil is visible in the site glass. This should be removed from time to time without stopping the mill by opening the needle valve below the sight gauge.

Loss of oil is shown the scale at the ram end of the accumulator; if 75% of the oil is lost then the ram trips the limit switch to operate an alarm signal.

The control panel is connected to the cylinders by means of a hydraulic ring main. The gas line has standard bulkhead fitting so that when the line is broken the pressure is released to operate pressure switch isolating the mill electrically.

To isolate the cabinet without losing all the gas pressure, a needle valve is fitted just above the accumulator.

8.15.2 OPERATION & COMMISSIONING OF LOADING PANEL

Ensure that all "**Normally open**" valves are open and that all " **Normally closed**" valves are closed. These are shown in the drawing no. U2908 34 00 - 3.

Close the regulating valve and connect the gas bottle. The regulating valve is a diaphragm valve. By screwing the knob, the valve opens and the screw it out the valve closes.

Open the gas bottle valve to check the precharge pressure. It should be approx. 150 Kg/cm^2 for fully charged bottle. Open the regulating valve to apply a gas pressure of approximately 15 Kg/cm^2 to the system; that the accumulator ram should extend. If it does not extend, then open the needle valve that is in parallel with the oil relief valve to allow the ram to move until it coincides with the mark "**D**" operate the hand pump to bring it back to the point **D.**

The regulating valve can now be re-adjusted to give the required system pressure. If, while doing this, the precharge pressure falls below the minimum precharge pressure, close the needle valve above the gas relief valve, change the bottle with a new one and then open the needle valve again.

COUTION : ONLY NITROGEN GAS IS TO BE USED

8.15.3 NORMAL OPERATION:

The position of the accumulator ram and the sight gauge should be checked daily as these will both indicate loss of oil. Never allow the oil level in the tank to fall below half fill. Top up with fresh, clean oil, not forgetting to replace the cap afterwards to prevent the ingress of coal dust.

8.15.4 SHUT DOWN

To shut down the system completely, slowly open the needle valve in parallel with the oil relief valve. This will allow the gas pressure to force the oil in the accumulator, back to the supply tank. Close the valves on the gas bottle than open the valve below the sight gauge; this will relieve all the gas pressure in the system. This is required only when the mill has to be stopped for extended periods or, if work is to be taken on the loading unit, requiring releasing the force on the grinding elements.

COUTION : DO NOT SHUT DOWN THIS SYSTEM WHILE THE MILL IS RUNNING

8.15.5 MAINTENANCE

The importance of maintaining a high degree of cleanliness while working on this hydraulic system cannot be over emphasised. Because of the close tolerances involved in cylinders and valves, it is imperative that the system remains completely free from contamination that might cause malfunctioning or failure of these components. To prevent dirt from entering the system, the following precautions must be taken before any work is commenced.

- Thoroughly clean the outside of fittings, pipe components and the surrounding structure before disconnecting any pipes.
- When the pipelines are disconnected, fit dust caps to the pipes and component unions, mask any exposed surfaces immediately and make certain that it is impossible for foreign matters to enter the system. If suitable capping cannot be achieved by use of cap or

plugs, place a vinyl plastic bag over the pipe or connections, securing it with an elastic band.

- When drying pipes or components use only compressed air that has been filtered and dehydrated; that is air from a high pressure bottle, **not from local compressor.**
- When flushing pipes, use clean carbon tetrachloride from a cleaner.
- Maintain clean hands and tools when working on the hydraulic system.

Keeping the pipelines and components clean will ease the locating external leakages. In some cases leakage may occur at a pipe coupling that has a metal to metal joints this may possibly be reminded by tightening the tube nuts, but care must be taken to avoid over tightening. If normal tightening is ineffective, examine the parts of the joint and replace as necessary. Teflon tapes could be used with the screwed fittings to stop leakages.

9.0.0 COMMISSIONING

9.1.0 PRE COMMISSIONING CHECKS

1. Foundation:

- Remove all foreign material.
- Check all the foundation bolts of the mill tighten them, if necessary. Normally, this may not be required.

2. Gearbox and motor:

- Check the pads of the gearbox. These pads will take the total load of the mill. Hence these pads bear well more than 80% seating on the base plate. Check with filler gauge. 0.05 mm thick filler gauge should not pass.
- Check the oil level in the gearbox. Fill correct grade oil and up to correct level.
- Check that all the holding down bolts are tightened properly with spring washers.
- Check the mill motor is firmly erected and correctly aligned to the gearbox coupling.
- Check all motors for correct direction of rotation.
- Check that gearbox is located in its position.
- Check that the bolts between the gearbox output coupling and yoke are to be properly tightened.
- Measure the gap between the yoke and air seal. The gap should be uniform and approximate 0.75 mm.
- Check all electrical interlocks and alarms.

3. Check the mill from inside.

- Check that the free throat area is correct.
- Check that scrapper ploughs are firmly secured.
- Check that spider guide clearance is correct.
- The push rod of the loading unit should be properly sitting in the cup and swivel pin is engaged in the slot of the cup.

4. Check that loading cylinders are properly secured to the cylinder support bracket. Check the tightness of support bracket holding bolts.

5. Check that the relief gate operates freely and pyrite box door open and close without difficulty.
6. Check the setting of the classifier vanes. Refer LNV drawing. Grit cone to be bolted securely to classifier stator.
7. Check the position of airflow measuring instrumentation.
8. Check leakage of gas and oil and if required take remedial action after pressurizing loading units.
9. Check that all pipelines gas and oil are connected properly to cylinder.

9.2.0 REGULAR CHECKS – AT LONGER INTERVALS.

Grinding Elements Maintenance.

The maximum economic life of grinding elements can only be achieved by adhering to an inspection and maintenance schedule, which implies keeping records datewise:

- hours of operation
- Average mill capacity
- Ball diameter
- Ring profile and minimum thickness
- Load per ball.

Factors affecting wear for a given material being ground is: -

a) Running Hours:

Generally the rate of wear will be greater when the mill is on low capacity than on high capacity, because of greater metal to metal contact. Hence mills should be operated near their rated output to minimise grinding element wear rate.

b) Minimum rate of wear occurs when the contour of ring and ball is closely matched. While this would indicate that it is preferable to wear the first set of balls down to discharge size, two factors have also to be considered. Firstly, a pair of rings will usually outlast two sets of balls and, secondly, the gap between first and last ball, when all are touching, increases with wear. Hence a system of ball rotation is recommended, including the addition of a fill-in ball.

10.0.0 STARTING THE MILL

After completion of pre-commissioning checks and receiving clearance from electrical department, start cooling water supply to gearbox. Next start the mill gearbox lubrication equipment and wait for 5 minutes so that lubrication oil will reach all the bearings etc.

Then start the classifier motor. After 1-2 minutes start the mill drive motor to spin the mill. This is preliminary start-up. Observe the mill for any noise or unnatural sound.

STOP

WARNING: DO NOT RUN THE MILL MORE THAN 5 MINUTES WITHOUT MATERIAL.

Before mill is commissioned with material, the calibration of primary air measuring equipment and the raw coal feeder is absolutely necessary.

Once the mill is stabilised and achieved rated capacity of production check the fineness of coal.

11.0.0 LUBRICATION

The following are the recommended lubricants. However, equivalent of the other manufacturer lubricants can also be used.

Sl.no.	Parts to be lubricated	Recommended lubricant
1	Gearbox	Oil lubrication • ISO-VG-320 oil • IPOL compound – 460 • Parthan EP VG – 460 or equivalent • 1700 Liters for 1st filling
2	Gear box seals	Grease lubrication High drop point grade EP2
3	Geared coupling	Grease lubrication • Servoplex EP1 - IOCL or • Alvamia EP2 – SHELL • Frequency : 2-3 times in a week
4	Loading cylinder	Shell gear oil SAE 80 or equivalent
5	Motor bearing	Grease Servogem 3 or equivalent

12.0.0 LIST OF ERECTION DRAWINGS

SL. No	Description	Drawing No.
1	General arrangement of BR Mill – sectional view	ECM-000255-1006
2	Foundation load data	ECM-000255-1002
3	Load bearing packers	C-BR-1217
4	Extension rail with withdrawal trolley	D-BR-1344
5	Drive arrangement	D-BR-1344
6	Drive motor – CUSTOMER SCOPE supply	
7	Resilient coupling	SIZE:- ET2200
8	Combined Bevel helical Gearbox	6003859/1
9	Lubrication system with gearbox	6003847
10	Arrgt of gearbox base plate	D-BR-1202
11	Mill housing support	D-BR-1179
12	Mill housing	D-BR-1187
13	Arrgt of reject box	C-BR-1258
14	Assembly of relief gate	C-BR-1228
15	Ball loading assembly	D-BR-1194
16	Push rod assembly	D-BR-1239
17	Arrgt of air seal housing at yoke	C-BR-1220
18	Spider Guide Arrangement	D-BR-1243
19	Arrangement of Friction grip bolts	B-BR-1214
20	Lower classifier Assembly	D-BR-1226
21	Classifier general arrangement	CLF13-1.1638
22	Schematic arrgt of loading system	U2908 30 00-3

13.0.0 WEIGHTS OF MILL COMPONENTS

The following are approximate weight of components for erection purpose only. Suitable marginal weight to be added while selecting lifting slings and shackles.

SL. No.	Description	Qty	Weight in Kg
1	Drive motor	1	8,500
2	Coupling between motor and gearbox	1	200
3	Gearbox • With oil • Without oil	1	 14,920 14,520
4	Mill housing support	1	19,553.5

5	Arrgt of gear base plate	1	4,250
6	Mill housing + ball loading door+ relief gate + throat assembly	1	25,000
7	Spider arm	1	140
8	Reject box with cut-off gate	1	376.5
9	Yoke	1	26,150
10	Bottom grinding ring	1	13,660
11	Single ball • Regular • Fill in ball	 1 1	 2,200 1,452
12	Top grinding ring	1	14,960
13	Spider assembly	1	17546.2
14	Spider guide assembly	1	2,019
15	Throat plate – in segments	1 set	4,200
16	Ball loading assembly	1	1,725
17	Loading cylinder	1	150
18	Lower classifier housing	1	13,912
19	Lower casing	1	1,866
20	Grit cone	1	3,256
21	Roof casing	1	3,144
22	Upper casing	1	2,683
23	Rotor assembly	1	1,604
24	Classifier drive assembly	1	1,087
25	Pocket assembly	1	2,883
26	Bearing cartridge and tie rod assembly	1	998

14.0.0 SPARES

The following are the recommended critical spares for two years operation maintenance. Spares for dynamic classifier refer M/s. LNV's operation and maintenance manual.

Sl.No	Description	Quantity
A	**Spares for Gearbox**	
1	Bevel gear pair	1 set
2	Bearing isolator	1 no.
3	seal	1 no.
B	**Spares for Mill**	
1	Grinding elements	1 set
2	Scraper assembly	2 nos.
3	Spider Guide Wear Plates	1 set
C	**Spares for Loading unit assembly**	
1	Seal for cylinder	8 nos.
2	Bellows for cylinder	4 nos.
3	Needle valve	4 nos.

4	Accumulator	1 no.
5	Non return valve – gas	1 no.
6	Non return valve - oil	1 no.

15.0.0 ALLOWABLE DEVIATIONS

Sl.No.	Detail	Allowable deviations
1	Centerline of one foundation bolt to another foundation bolt	± 0.5 mm
2	Diagonal of one foundation bolt to another foundation bolt	± 1.0 mm
3	Variation in the grouting margin ♦ Up to 50 mm ♦ More than 50 mm	 ± 5.0 mm ± 10.0 mm
4	Leveling of mill components ♦ Up to 2 Mts. ♦ 2 Mts. to 5 Mts.	 0.2 mm/m 0.3 mm/m
5	Permissible tolerance for level of pads on which Gear box sits.	+0.10 mm
6	Level of the grout from top of base plate	Max: Flush Min: - 6 mm
7	Clearance between I/D of air seal housing and O/D of air seal ring	H7/f8
8	Clearances between I/D of seal ring and O/D of Yoke stem.	0.8 mm
10	Gap between gear base pad with gearbox.	0.05 mm filler gauge should not pass
11	% Of blue matching required between packer of mill housing support and base plate.	All packers should be tight and tack welded to each other after final levelling. Blue matching not required.
12	% Of contact between gearbox base plate pads with gearbox.	80% blue matching contact.
13	Leveling value with permissible limit of gearbox & other mill components.	Levelling will be checked with Master level. Allowable limit for levelling will be 0.05 to 0.10 mm Max.

14	Levelling of yoke flange	Both inner and outer faces to be in one plane. Allowable out of flatness will be > 0.075 mm
16	Value with permissible limit of outer and inner landing level of top & bottom grinding ring.	Blue matching of bottom ring and yoke flange and top ring & spider. **A).** Outer landing face: There should be at least 80% contact surface and length of non-contact portion should not be more than 300 mm at one stretch. **B).** Inner landing will be lower than outer landing by 0.1-mm max. Its level should not effect blue matching of outer landing.
17	Permissible limit in dia. of 10 nos. (one set) of grinding ball.	The variation in diameter of one set ball - 2-5 mm.
18	Throat gap value is required.	Initially the gap may be set around 10-14 mm and would be set as per requirement during operation

16.0.0 LIST OF TOOLS AND TACKLES

The following are the special and general tools and tackles required at site for erection of 120(10) Ball race mill

Sl.no.	Description		Quantity
1	Crawler crane – 50 ton capacity	:	1 no.
2	Power winch with 100 Mts. wire rope 25 mm dia – 25 Ton capacity	:	1 no
3	Chain pulley block	:	
	♦ 10 ton capacity		2 no
	♦ 5 ton capacity		2 nos
	♦ 2 ton capacity		2 nos
4	Hydraulic jack	:	
	♦ 10 ton capacity		3 nos
5	Misc. items	:	
	♦ Wire rope slings		One lot containing
	♦ Shackles		all sizes
	♦ U clamps		
	♦ Manila ropes		
	♦ Crow bar		
	♦ Turn buckles		
6	Fitter's tool sets one set in each group of workers	:	1 sets
7	Wrenches for nuts size up to M48 including torque wrench.	:	1 set
8	Hammer of all sizes including sledge hammer, brass hammer and rubber hammer	:	1 sets
9	Filler gauges	:	
	♦ 300 mm long		1 no
	♦ 100 mm long		1 no
10	Dial gauge indicator – in mm scale	:	2 nos.
11	Precision master level	:	
	♦ 300 mm long		1 no
	♦ 150 mm long		1 no
12	Surveying instruments	:	
	♦ Engineers level – Precision		1 no
	♦ Theodolyte – Precision		1 no
13	Shims	:	As required
	♦ MS plate pieces of 1 mm to 50 mm –		

	all sizes ♦ Brass shims of thickness below 1 mm – all sizes		
14	Heavy section MS channels, beams, seasoned wooden sleepers for mill shell dragging	:	One lot as per site condition
15	Channels, angles etc. for temporary scaffolding etc., required during erection at site	:	One lot as per site condition
16	Gas welding/Gas cutting and electric welding sets.	:	1 set
17	Portable hand grinder, angle grinder/Disc grinder and flexible shaft grinder,	:	1 no.
18	Transparent plastic tube for water level 12 mm dia.	:	50 m
19	Scrapping tools	:	1 set
20	Adhesive pastes ♦ Hold tight ♦ Loctite - 406	:	2 tube each
21	Prussian blue color	:	8 tubes
22	Straight edge	:	1 no.
23	Slip gauge	:	1 no.
24	Magnetic center	:	1 no.
25	Hand lamp	:	2 nos
26	Oil ▪ Lubricating oil ▪ Flushing oil	:	One drum each
27	Grease	:	2 kg.
28	Sand paper – all size	:	One box
29	Metric threads tap set	:	One
30	Plumb	:	1 sets
31	All other general tools and tackles not mentioned in the above list but generally used in erection of mechanical equipment	:	2 sets
32	First aid Box	:	1 set

Nitrogen filling set
Torque tensioner
Tummy bar For rotation of gear box